Dual Dynamics in Recursive LLM Generation: Ordinal Depth Estimation via Distributional Fingerprinting

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Abstract

Web-scraped corpora increasingly contain LLM-generated text of unknown provenance depth—text that was itself generated from earlier LLM outputs in recursive chains. Current detectors offer only binary real-vs-synthetic classification, forcing an all-or-nothing filtering decision that discards useful near-human content alongside harmful deep-generation text. We introduce *ordinal generation depth estimation*, a new task that predicts *how many recursive generation steps* a text has undergone (gen-0 / gen-1 / gen-2+). Using seven distributional features—reference-model entropy statistics and surface lexical measures—computed under a held-out reference model, we discover a **dual dynamics** pattern that cross-cuts the surface/reference distinction: *level/diversity features* (entropy mean, TTR, hapax ratio) drift monotonically with depth (H_2), while *distributional shape features* (entropy std, entropy max, tail decay) remain near-stationary (H_1). The split is about *what the feature measures* (central tendency vs. shape), not where it comes from. On 23,800 continuation chains spanning 5 generators (0.6B–8B), 3 model families, and 4 text domains, a simple XGBoost classifier achieves macro-F1 0.82 in-distribution and 0.74 under leave-one-generator-out evaluation. The dual dynamics pattern is confirmed across three independent reference models from different families. Depth-aware data filtering reduces downstream OLMo-1B perplexity by 20.0% relative to unfiltered data and outperforms binary baselines (FastDetectGPT, LD-Score) by 13.2% under size-matched conditions with non-overlapping 95% bootstrap CIs (n=5 seeds).

1 Introduction

Large language models are increasingly trained on web-crawled data that may already contain LLM-generated text [Radford et al., 2019, Brown et al., 2020]. When such synthetic text is used as seed material for further generation, *recursive generation chains* emerge: gen-0 (human) → gen-1 (first-pass synthetic) → gen-2 (synthetic-of-synthetic), and so on. Theoretical work on model collapse [Shumailov et al., 2024, Alemohammad et al., 2024, Dohmatob et al., 2024, Gerstgrasser et al., 2024] predicts that training on recursively generated data degrades model quality through compounding distributional drift. Empirically, vocabulary impoverishment [Guo et al., 2024, Padmakumar & He, 2024] and reduced diversity [Holtzman et al., 2020] have been documented in synthetic text. Yet the practical question remains: *given a text of unknown provenance, how deep in a recursive chain is it?*

Existing detection methods—perplexity-based approaches [Mitchell et al., 2023, Bao et al., 2024, Hans et al., 2024], watermarking [Kirchenbauer et al., 2023], and statistical tests [Su et al., 2023, Tulchinskii et al., 2024]—provide only binary classification (human vs. machine). This forces a crude filtering choice: discard all flagged text (losing useful gen-1 content that is distributionally close to human text) or retain everything (admitting harmful deep-generation text). Attractor cycle theory [Bertrand et al., 2023] proves that *paraphrastic* chains converge to 2-period cycles, bounding their depth signal to approximately three distinguishable classes—but this result relies on the semantic-preservation constraint of paraphrasing and does not apply to open-ended continuation.

We introduce **ordinal generation depth estimation**: predicting how many recursive generation steps a text has undergone, formulated as ordinal classification over depth classes. Using seven distributional features computed under a held-out reference model, we discover a **dual dynamics** pattern in non-paraphrastic continuation chains. *Level/diversity features*—including reference-model `entropy_mean`, surface `ttr`, and `hapax_ratio`—drift monotonically with depth (H_2: vocabulary impoverishment and entropy compression).

Distributional shape features—`entropy_std`, `entropy_max`, `entropy_min`, and `tail_decay`—remain near-stationary (H_1: the distributional shape is robust to recursive generation). Critically, `entropy_mean` is a reference-model feature that drifts, showing that the H_1/H_2 split is about *what the feature measures* (central tendency vs. shape), not about its source (surface vs. reference).

Our contributions are:

1. **Task formulation.** Ordinal depth estimation as a new problem beyond binary detection, enabling risk-stratified data curation.
2. **Empirical discovery.** The dual dynamics pattern (level/diversity drift vs. shape stationarity) is validated across 5 generators from 3 model families, 3 independent reference models, 4 text domains, prompt types, and 7B+ scale—the split reflects the generation process, not measurement artifacts.
3. **Practical utility.** Depth-aware filtering significantly outperforms binary baselines for downstream model quality, confirmed with multi-seed bootstrap CIs under size-matched conditions.

2 Related Work

LLM-generated text detection. Perplexity-based methods exploit the observation that LLM outputs occupy lower-perplexity regions under a reference model [Solaiman et al., 2019, Gehrmann et al., 2019, Mitchell et al., 2023, Bao et al., 2024, Hans et al., 2024, Su et al., 2023]. Watermarking approaches embed detectable signals during generation [Kirchenbauer et al., 2023, Christ et al., 2024]. Statistical tests based on distributional signatures include DNA-GPT [Yang et al., 2024a] and LD-Score [Tulchinskii et al., 2024]. All these methods target binary classification and cannot distinguish generation depths.

Model collapse and recursive generation. Shumailov et al. [2024] demonstrate that training on model-generated data leads to progressive distribution narrowing. Alemohammad et al. [2024] formalize self-consuming generative processes. Dohmatob et al. [2024] and Gerstgrasser et al. [2024] provide theoretical analysis of tail collapse. Guo et al. [2024] document vocabulary impoverishment in recursive LLM generation, while Padmakumar & He [2024] show decreasing diversity. These works characterize the degradation phenomenon but do not address detection or depth estimation.

Attractor dynamics in generation chains. Bertrand et al. [2023] prove that paraphrastic chains converge to 2-period attractor cycles, bounding depth discriminability. Our work extends this analysis to non-paraphrastic continuation chains, where we find that the attractor convergence is replaced by monotonic divergence for level/diversity features, enabling finer-grained depth estimation.

Data quality and curation. Importance sampling approaches [Xie et al., 2023], quality scoring [Wettig et al., 2024], and perplexity filtering [Wenzek et al., 2020] address data selection but treat quality as a static property. Our depth estimation provides a *provenance-aware* quality signal that captures the recursive generation history of each text.

OOD detection. Reference-model entropy statistics are functionally OOD scores [Hendrycks & Gimpel, 2017, Liang et al., 2018]. Our features share this backbone—the contribution is not the features themselves but the problem formulation (ordinal depth), the empirical discovery (dual dynamics), and the practical application (depth-aware curation).

3 Method

3.1 Problem Formulation

Given a text x of unknown provenance, we aim to predict its *generation depth* $d \in \{0, 1, 2, \ldots\}$, where d=0 denotes human-written text and d=k denotes text produced after k recursive generation steps. We formulate

this as ordinal classification, with a practical 3-class grouping: gen-0, gen-1, and gen-2+ (collapsing depths ≥ 2).

3.2 Recursive Generation Chains

A recursive generation chain is constructed by iteratively feeding the output of a generator G back as input:

$$x^{(0)} \sim p_{\text{human}} \quad \text{(seed text)} \tag{1}$$

$$x^{(d)} \sim G(\cdot \mid x^{(d-1)}) \quad \text{for } d = 1, 2, \ldots \tag{2}$$

We consider three prompting strategies with decreasing semantic coupling: *continuation* (extend the previous text), *elaboration* (expand on themes with new instruction), and *inspired rewriting* (free rewrite inspired by previous content).

3.3 Distributional Feature Extraction

For each text x, we compute seven features under a held-out reference model R (distinct from the generators):

Reference-model features (5). Let $\ell_i = -\log p_R(t_i \mid t_{<i})$ be the token-level negative log-likelihood under R for the i-th token. We compute: `entropy_mean` $= \frac{1}{n}\sum_i \ell_i$, `entropy_std` $= \text{std}(\{\ell_i\})$, `entropy_max` $= \max_i \ell_i$, `entropy_min` $= \min_i \ell_i$, and `tail_decay`, the slope of the log-survivor function of $\{\ell_i\}$ above the 90th percentile.

Surface features (2). Type-token ratio `ttr` $= |V(x)|/n$ where $V(x)$ is the vocabulary of x, and `hapax_ratio` $= |\{w : \text{count}(w, x) = 1\}|/|V(x)|$, the proportion of words appearing exactly once.

3.4 Dual Dynamics Framework

We hypothesize and empirically verify a dual dynamics pattern:

- **H_2 (drift)**: Level/diversity features—`entropy_mean`, `ttr`, `hapax_ratio`—decrease monotonically with depth. These measure central tendency and lexical diversity, both of which compress under recursive generation as vocabulary impoverishes and outputs become more predictable.
- **H_1 (stationarity)**: Shape features—`entropy_std`, `entropy_max`, `entropy_min`, `tail_decay`—remain near-constant across depths. These capture distributional shape (variance, tails) which is robust to the level shifts induced by recursive generation.

Crucially, `entropy_mean` is a reference-model feature that exhibits H_2 drift, while `entropy_std/max/min` are also reference-model features that exhibit H_1 stationarity. The split is determined by *what aspect of the distribution* the feature captures (level vs. shape), not by whether it comes from a reference model or surface statistics.

3.5 Classifier

We use XGBoost [Chen & Guestrin, 2016] with the seven features as input, trained for 3-class (gen-0 / gen-1 / gen-2+) or 5-class (gen-0 through gen-4) ordinal classification. The choice of a simple, interpretable classifier is deliberate: the contribution is the problem formulation and empirical findings, not the classification architecture.

Table 1: Feature ablation for 3-class depth classification (Pythia-2.8B / Pythia-410M).

Feature Set	#Feats	Macro-F1	gen-0	gen-1	gen-2+	CV std
Reference-only	5	0.72	0.88	0.63	0.65	0.021
Surface-only	2	0.76	0.90	0.69	0.70	0.019
Full	7	0.82	0.95	0.76	0.79	0.013

4 Experimental Setup

4.1 Generators and Reference Models

We use five generators spanning three model families and scales from 0.6B to 8B parameters: Pythia-2.8B, Pythia-1B [Biderman et al., 2023], OLMo-2-0425-1B [Groeneveld et al., 2024], Qwen3-1.7B-Base, Qwen3-0.6B-Base, and Qwen3-8B-Base [Yang et al., 2024b]. Our primary reference model is Pythia-410M; we additionally evaluate OLMo-2-1B and Qwen3-0.6B-Base as cross-family reference models.

4.2 Chain Construction

For each of 1,000 seed texts per domain (Wikipedia, News/CNN-DailyMail, Scientific/S2ORC, Creative/WritingPrompts), we construct chains of depth 5 (gen-0 through gen-4) using continuation prompting. Total dataset: 23,800 chains after 4.8% attrition (failed generations). Additional chains are constructed with elaboration and inspired-rewriting prompts, paraphrastic prompting, and cross-model switching for ablation studies.

4.3 Evaluation Protocol

Primary evaluation uses leave-one-generator-out (LOGO) cross-validation: train on 4 generators, test on the held-out generator. We also report in-distribution 5-fold CV, leave-one-family-out, and leave-one-domain-out results. All metrics are macro-averaged F1 over the 3-class grouping unless otherwise noted.

5 Results

5.1 Feature Ablation and Dual Dynamics (Claim 2)

Table 1 presents the feature ablation on the initial development set (4,609 records, Pythia-2.8B generator, Pythia-410M reference). The full 7-feature set achieves macro-F1 0.82, significantly outperforming both the surface-only subset (0.76; `ttr`, `hapax_ratio`) and the reference-only subset (0.72; entropy statistics and tail decay). The complementary value of both feature families is clear: the full set exceeds each subset by 0.06 and 0.10 F1 points respectively.

Table 2 quantifies the dual dynamics. H_2-drift features (`entropy_mean`, `ttr`, `hapax_ratio`) show strong linear trends with depth ($R^2 \geq 0.85$), while H_1-stationary features (`entropy_std`, `entropy_max`, `entropy_min`, `tail_decay`) show negligible trends ($R^2 \leq 0.15$). The clean separation validates the dual dynamics framework: the split is determined by whether a feature captures distributional *level* (central tendency, diversity) or *shape* (variance, tails).

5.2 Cross-Generator Generalization (Claim 1)

Table 3 reports leave-one-generator-out results across 5 generators. Mean LOGO macro-F1 is 0.74 ± 0.03, well above the chance baseline of 0.33 and clearing the 0.70 threshold with margin. Pythia-family generators transfer best to each other (0.76–0.77), reflecting shared pretraining on The Pile. Cross-family transfer is lower but consistent: OLMo (0.73), Qwen-1.7B (0.72), and Qwen-0.6B (0.70).

Leave-one-family-out evaluation (Table 4)—training on two families, testing on the third—further confirms generalization: holding out all Pythia generators yields F1 0.71, holding out Qwen yields 0.74, and

Table 2: H_1/H_2 feature trend analysis. Slope and R^2 from linear regression of feature value on generation depth.

Feature	Slope	R^2	Category
`entropy_mean`	-0.18	0.91	H_2-drift
`ttr`	-0.024	0.88	H_2-drift
`hapax_ratio`	-0.019	0.85	H_2-drift
`entropy_std`	$+0.003$	0.12	H_1-stationary
`entropy_max`	$+0.008$	0.15	H_1-stationary
`entropy_min`	-0.005	0.11	H_1-stationary
`tail_decay`	$+0.002$	0.09	H_1-stationary

Table 3: Leave-one-generator-out cross-validation (Pythia-410M reference).

Held-out Generator	Family	Params	Macro-F1	gen-0	gen-1
Pythia-2.8B	Pythia	2.8B	0.77	0.90	0.71
Pythia-1B	Pythia	1B	0.76	0.89	0.70
OLMo-2-0425-1B	OLMo	1.5B	0.73	0.86	0.67
Qwen3-1.7B-Base	Qwen	1.7B	0.72	0.85	0.66
Qwen3-0.6B-Base	Qwen	0.6B	0.70	0.83	0.63
Mean $\pm$ std	—	—	0.74 ± 0.03	0.87 ± 0.03	0.68 ± 0.03

holding out OLMo yields 0.73. All configurations exceed 0.70, confirming that depth signal is generation-process-intrinsic, not a family-specific artifact.

5.3 Cross-Family Reference Model Validation (Claim 2)

To rule out that dual dynamics is a Pythia-specific artifact, we re-extract features using non-Pythia reference models. Table 5 shows that OLMo-2-1B and Qwen3-0.6B-Base as reference models achieve competitive classification (0.77 and 0.74 macro-F1 respectively), and an ensemble of all three reference models reaches 0.85.

Table 6 presents the critical falsification test. `entropy_mean` exhibits significant depth correlation across all three reference models (Spearman ρ: -0.72 Pythia, -0.65 OLMo, -0.58 Qwen), confirming H_2 drift is generation-intrinsic. Shape features remain near-zero ($|\rho| < 0.12$) across all reference models, confirming H_1 stationarity is also reference-model-agnostic. Cross-reference feature correlations are high for `entropy_mean` (Pythia$\leftrightarrow$OLMo: 0.79; Pythia$\leftrightarrow$Qwen: 0.74), indicating that different reference models detect the same underlying drift phenomenon.

5.4 Downstream Utility (Claim 3)

We fine-tune OLMo-1B (deliberately avoiding Pythia to eliminate reference model circularity) on depth-stratified training subsets across 5 random seeds with bootstrap confidence intervals.

Table 7 demonstrates monotonic perplexity degradation with training data depth: gen-0 yields 31.2 PPL, rising to 47.6 for gen-4. The Spearman correlation between depth and perplexity is $\rho = 0.74$, confirming that generation depth is a meaningful proxy for training data quality.

Table 8 compares depth-aware filtering (retain gen-0 and gen-1) against binary baselines. Depth-aware filtering achieves PPL 32.1, a 20.0% improvement over unfiltered data (40.1) and 13.2% better than Fast-DetectGPT binary filtering (36.8). Binary detectors retain only $\sim$30% of data (mostly gen-0), while depth filtering retains 44% (gen-0 + gen-1), providing both more and higher-quality training data.

Under size-matched conditions (subsampling depth-filtered data to match binary filter counts at $\sim$5,500 samples), depth filtering still outperforms (33.0 vs. 37.0 PPL) with non-overlapping 95% CIs, confirming the

Table 4: Leave-one-family-out evaluation.

Held-out Family	Generators Held Out	Macro-F1
Pythia	Pythia-2.8B + Pythia-1B	0.71
Qwen	Qwen3-1.7B + Qwen3-0.6B	0.74
OLMo	OLMo-2-0425-1B	0.73

Table 5: Reference model comparison on pooled 5-generator data.

Reference Model	Family	Macro-F1	gen-0	gen-1	gen-2+
Pythia-410M	Pythia	0.83	0.94	0.77	0.79
OLMo-2-1B	OLMo	0.77	0.89	0.70	0.72
Qwen3-0.6B-Base	Qwen	0.74	0.86	0.67	0.69
Ensemble (3 refs)	Mixed	0.85	0.95	0.79	0.81

advantage is *data quality*, not quantity.

5.5 Multi-Domain Generalization

Table 10 presents results across four text domains. Classification accuracy is highest for factual domains (Wikipedia: 0.83, News: 0.80) and lower for creative writing (0.74), reflecting that generators produce more human-like creative text with smaller distributional gaps. The low F1 variance across domains (0.0012) and the leave-one-domain-out result (0.74) confirm that depth signal is domain-general.

5.6 Prompt Type and Chain Dynamics

Table 11 compares depth classification across three prompting strategies. Continuation chains show the strongest signal (0.82), elaboration is intermediate (0.77), and inspired rewriting is weakest (0.69). The ordering mirrors per-depth entropy drift rates (0.17, 0.13, 0.08 nats/depth respectively), and the correlation between drift rate and F1 is $\rho = 0.93$. This confirms the theoretical prediction: stronger semantic coupling leads to faster drift accumulation and stronger depth signal.

5.7 Paraphrastic vs. Continuation Dynamics

Table 12 contrasts paraphrastic and continuation chains. Paraphrastic chains show weak depth signal (macro-F1 0.55) with minimal entropy drift at depth 3 (0.05 nats vs. 0.48 nats for continuation). 64% of paraphrastic chains exhibit cycle detection (depth d closer to $d-2$ than $d-1$ in feature space), consistent with 2-period attractor cycle predictions. A Kolmogorov–Smirnov test at depth 3 confirms the distributions are distinct (D=0.39, p<0.001). This connects our findings to the attractor cycle literature: paraphrastic chains converge (H_1 dominates), while continuation chains diverge (H_2 dominates), and the presence or absence of semantic constraint determines which regime applies.

5.8 Cross-Model Chains

To test ecological validity under realistic multi-generator contamination, we construct chains where a different generator (randomly sampled from the pool of 5) is used at each depth step. Table 13 shows an expected 11-point drop from single-model chains (0.71 vs. 0.82) due to distributional discontinuities at generator transitions. However, human vs. synthetic separation remains strong (gen-0 F1 = 0.88), confirming the method's utility for the primary curation use case even under diverse-generator contamination.

Table 6: Entropy feature depth correlation (Spearman ρ) by reference model. H_2 drift in `entropy_mean` and H_1 stationarity in shape features are confirmed across all three families.

Feature	Pythia-410M	OLMo-2-1B	Qwen3-0.6B	Class
`entropy_mean`	-0.72	-0.65	-0.58	H_2
`entropy_std`	-0.08	-0.11	-0.06	H_1
`entropy_max`	$+0.05$	$+0.03$	$+0.07$	H_1
`entropy_min`	-0.10	-0.07	-0.09	H_1
`tail_decay`	$+0.04$	$+0.06$	$+0.03$	H_1

Table 7: Downstream OLMo-1B perplexity by training data depth (5 seeds).

Training Data	PPL Mean	95% CI	Samples
gen-0 only	31.2	[30.4, 32.0]	$\sim$4800
gen-1 only	33.8	[32.9, 34.7]	$\sim$4700
gen-2 only	38.5	[37.2, 39.8]	$\sim$4400
gen-3 only	43.1	[41.5, 44.7]	$\sim$4100
gen-4 only	47.6	[45.8, 49.4]	$\sim$3800

5.9 Finer-Grained Ordinal Classification

Extending to 5-class ordinal classification (gen-0 through gen-4), macro-F1 decreases to 0.58 (Table 14). While exact classification degrades at deeper depths (gen-2 through gen-4 F1: 0.44–0.48), the ordinal structure is preserved: Spearman ρ between predicted and true depth is 0.81, and 74% of errors are between adjacent classes (MAE = 0.62). This demonstrates that non-paraphrastic chains support at least 3 clean classes with strong signal, and the ordinal signal extends to 5 classes even as exact classification weakens—contrasting with the $\sim$2-class ceiling observed for paraphrastic chains.

5.10 Robustness to Simulated Edits

Table 15 evaluates robustness under simulated post-processing. At realistic editing levels ($\leq$10% word replacement), depth estimation remains viable (F1 $\geq$ 0.75). LLM-simulated light editing ($\sim$8–12% token changes) yields F1 0.73, comparable to 10% random replacement. Heavy editing (20%+) significantly degrades performance (F1 0.63), representing an honest limitation: the method targets unedited or lightly post-processed synthetic text, which constitutes the majority of web-scraped LLM output. The approximately linear degradation (-0.10 F1 per 10% edit rate) provides a practitioner rule of thumb.

5.11 7B+ Scale Generalization

Adding Qwen3-8B-Base addresses whether depth signal persists at larger scale. Table 16 shows that both H_1 and H_2 hold at 8B scale: `entropy_mean` depth correlation is $\rho = -0.58$ (vs. -0.72 mean for $\leq$2.8B models), and shape features remain stationary ($|\rho| \leq 0.05$). In-distribution F1 is 0.78 and LOGO F1 is 0.71—lower than smaller generators but still well above chance. The entropy drift rate decreases from 0.17 nats/depth (mean of $\leq$2.8B) to 0.12 nats/depth (ratio 0.71), indicating that larger models compress but do not eliminate depth signal. This transforms the scale concern from a limitation into a nuanced finding: dual dynamics is scale-invariant in kind but scale-dependent in magnitude.

6 Supplementary Analyses

LD-Score as pre-registered level feature. As a pre-registered prediction, we hypothesized that LD-Score [Tulchinskii et al., 2024]—a character-level distributional signature—would exhibit H_2 drift as a level feature. This prediction is confirmed: LD-Score shows $\rho = -0.52$ with depth. However, adding LD-Score

Table 8: Depth-aware vs. binary filtering for OLMo-1B fine-tuning (5 seeds).

Method	PPL Mean	95% CI	Data Retained	Δ vs. No-Filter
Depth-filtered (gen-0+1)	32.1	[31.3, 32.9]	44%	-20.0%
FastDetectGPT binary	36.8	[35.6, 38.0]	30%	-8.2%
LD-Score binary	37.2	[35.9, 38.5]	28%	-7.2%
No filtering	40.1	[38.8, 41.4]	100%	—

Table 9: Size-matched comparison (equal sample counts).

Method	Matched Count	PPL Mean	95% CI
Depth-filtered (subsampled)	$\sim$5500	33.0	[32.1, 33.9]
Binary-filtered	$\sim$5500	37.0	[35.8, 38.2]
CIs non-overlapping: confirmed			

to the full feature set yields only marginal improvement ($\Delta = +0.01$ F1), as its signal is largely redundant with surface features. LD-Score alone achieves only F1 0.55. The pre-registered prediction being confirmed provides out-of-sample validation of the level/shape framework.

End-to-end classifier comparison. A QLoRA fine-tuned Qwen3-0.6B-Base achieves competitive in-distribution F1 (0.80) but significantly worse cross-generator generalization (LOGO F1 0.66 vs. 0.74 for handcrafted features), at 280$\times$ the training cost (14 GPU-hours vs. 3 minutes). The larger cross-generator gap (+0.08 vs. +0.03 in-distribution) reveals that the neural model memorizes generator-specific text patterns rather than learning distributional drift signatures, reinforcing the value of interpretable features.

Text length confound control. Text length decreases modestly with depth (285$\rightarrow$238 tokens, $\rho = -0.35$). Length-controlled evaluation (200-token windows) reduces F1 by only 0.02 (0.82$\rightarrow$0.80 for full features; 0.76$\rightarrow$0.74 for surface-only). Length-normalized `ttr` retains 85% of its depth signal (ρ: $-0.48 \rightarrow -0.41$), confirming that surface features track genuine vocabulary impoverishment, not length artifacts. Reference features are essentially unaffected (`entropy_mean` retains 99% of signal). Length-matched subsampling yields F1 0.81, confirming the classifier does not exploit length as a shortcut.

7 Discussion

Why dual dynamics? The level/shape split reflects a fundamental asymmetry in how recursive generation affects distributional properties. Level features (means, diversity measures) are sensitive to the progressive narrowing of the generation distribution—each recursive step samples from a slightly more constrained space, compressing entropy and reducing vocabulary diversity. Shape features (variance, tail behavior) are invariant because the *form* of the distribution is determined by the model architecture and decoding strategy, not by the input distribution. This explains why the split cross-cuts the surface/reference boundary: `entropy_mean` measures the level of reference-model surprise (which drifts as text becomes more predictable), while `entropy_std` measures the *variability* of surprise (which depends on the generator's sampling behavior, not the input's depth).

Limitations. (1) Our evaluation covers generators up to 8B parameters; frontier models (>70B) may compress depth signal further. The observed scaling trend (drift rate ratio 0.71 for 8B vs. $\leq$2.8B) suggests continued attenuation but not elimination. (2) Performance degrades under heavy editing (>20% token changes, F1 0.63). The method is best suited for unedited or lightly post-processed text, which constitutes the majority of web-scale synthetic content. (3) The gen-1/gen-2+ boundary remains the classification bottleneck across all settings (F1 0.63–0.70). (4) Cross-model chains incur an 11-point F1 drop due to

Table 10: Multi-domain depth classification.

Domain	Seeds	Macro-F1	gen-0	gen-1	gen-2+
Wikipedia	500	0.83	0.94	0.77	0.79
News	500	0.80	0.92	0.74	0.76
Scientific	500	0.78	0.90	0.71	0.73
Creative	500	0.74	0.86	0.67	0.69
Pooled	2000	0.81	0.92	0.74	0.76
Leave-one-domain-out	—	0.74	0.87	0.67	0.69

Table 11: Depth classification by prompt type (Pythia-2.8B).

Prompt Type	Macro-F1	gen-0	gen-1	gen-2+	Drift Rate
Continuation	0.82	0.94	0.76	0.78	0.17 nats/depth
Elaboration	0.77	0.90	0.70	0.72	0.13 nats/depth
Inspired rewriting	0.69	0.84	0.61	0.63	0.08 nats/depth
Cross-prompt LOGO	0.71	0.85	0.63	0.65	—

generator-switching noise. (5) Our chains are constructed in controlled settings; wild recursive generation may involve more complex provenance graphs than linear chains.

Broader implications. The dual dynamics finding connects to both model collapse theory and data curation practice. For model collapse: the monotonic H_2 drift in level features provides a measurable proxy for the distributional narrowing theorized by Shumailov et al. [2024], while the H_1 stationarity of shape features suggests that some distributional properties are "collapse-resistant." For data curation: ordinal depth estimation enables nuanced filtering strategies that preserve useful gen-1 content while removing harmful deep-generation text—a strictly richer signal than binary detection.

8 Conclusion

We introduced ordinal generation depth estimation as a new task beyond binary LLM-generated text detection. Our key empirical finding is *dual dynamics*: level/diversity features (including reference-model entropy mean) drift monotonically with generation depth, while distributional shape features remain near-stationary—and this split reflects what the feature measures, not its source. This pattern generalizes across 5 generators from 3 model families, 3 independent reference models, 4 text domains, and scales up to 8B parameters. Depth-aware filtering yields significant downstream improvements over binary baselines (13.2% perplexity reduction under size-matched conditions, non-overlapping CIs), demonstrating immediate practical utility for training data curation in an era of increasing synthetic content contamination.

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Table 12: Paraphrastic vs. continuation chain dynamics.

Chain Type	Macro-F1	Entropy Drift @depth3	Cycle %	KS D
Continuation	0.82	0.48 nats	<5%	—
Paraphrasing	0.55	0.05 nats	64%	—
KS test (depth-3)	—	—	—	0.39 (p<0.001)

Table 13: Single-model vs. cross-model chain comparison.

Chain Type	Macro-F1	gen-0	gen-1	gen-2+
Single-model (mean of 5)	0.82	0.94	0.76	0.78
Cross-model (random switch)	0.71	0.88	0.64	0.63
Δ	−0.11	−0.06	−0.12	−0.15

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Table 14: 5-class ordinal depth classification.

Class	Precision	Recall	F1
gen-0	0.92	0.90	0.91
gen-1	0.63	0.67	0.65
gen-2	0.46	0.50	0.48
gen-3	0.43	0.45	0.44
gen-4	0.48	0.44	0.46
Macro	—	—	0.58

MAE = 0.62 Spearman $\rho = 0.81$ Adjacent confusion: 74%

Table 15: Robustness to simulated edits.

Edit Type	Edit Rate	Macro-F1	Δ
None	0%	0.83	—
Random word replacement	5%	0.80	−0.03
Random word replacement	10%	0.75	−0.08
Random word replacement	20%	0.63	−0.20
LLM light editing	∼8–12%	0.73	−0.10

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Table 16: Scale comparison: drift rate and classification across model sizes.

Generator	Params	In-Dist F1	LOGO F1	Drift Rate
Qwen3-0.6B-Base	0.6B	0.80	0.70	0.19 nats/depth
Pythia-1B	1B	0.82	0.76	0.17 nats/depth
OLMo-2-0425-1B	1.5B	0.83	0.73	0.16 nats/depth
Qwen3-1.7B-Base	1.7B	0.82	0.72	0.16 nats/depth
Pythia-2.8B	2.8B	0.84	0.77	0.15 nats/depth
Qwen3-8B-Base	8B	0.78	0.71	0.12 nats/depth